



I-STEM INTERNSHIP PROGRAM

Objectives

- ✓ To engage unemployed technical manpower with a Diploma/graduate degree in Science, Engineering and Technology
- ✓ To develop trained technical manpower to maintain and operate R&D facilities, that otherwise remain idle most of the time
- ✓ To train technical manpower to open-up, strip, repair, and/or retrieve useful spare parts from written-off equipment and maintain a library of such accessories at I-STEM portal
- ✓ To create a pool of such technical manpower trained to meet the need for equipment and maintenance engineers at R&D facilities across the country.
- ✓ To enhance the working life of sophisticated instruments by regular maintenance and thereby making (import) capital-expenditure more cost-effective.
- ✓ To enable optimal use of the high-end equipment and provide opportunities for host institutions to generate funds by sharing the equipment and by conducting skill development training using in-house experts (faculty members/technical staff/students).
- ✓ To fill the void to operate an instrument when trained technical manpower (e.g. Research/PhD candidate) leaves a facility, or promoted to another laboratory of importance, also to reduce dependency of facilities on research students.

Mechanism

- ✓ Host Institutions may provide the details of their requirements in prescribed format at the portal
- ✓ Advertisement of the internship program/requirement through I-STEM national portal
- ✓ Registration and Selection of interns on a regional basis in coordination with I-STEM team
- ✓ Interns will be paid a suitable stipend by I-STEM for a limited period
- ✓ After being trained, Interns may be offered employment by the host/different institution; or they may seek opportunity through I-STEM portal or practice on their own as Tools Consultant, via the I-STEM, at any of the labs listed on the portal across the country.

How to get engaged with various R&D Labs:

Interested candidates may apply to the internship program throughout the year via the I-STEM portal <https://www.istem.gov.in/employment-opportunities>.

Suitable candidates will be shortlisted from time to time and be asked to join the program as and when custodian institutions send request for an internship to the I-STEM team. For more details, please visit www.istem.gov.in



Dr. Sanjeev Kumar Shrivastava
National Coordinator, I-STEM
E-mail: sanjeevs@iisc.ac.in/ istem.india@gmail.com

www.istem.gov.in

Toll Free: 1800-425-3281

@Istem_IISc
@I-stem iisc



The water droplets on the touch screen add to the discomfort. This new readout methodology enables a much-improved sensitivity, ignoring any element (water or fabric) present between the finger and screen for a seamless operation.

It also enables the usage of low-cost printable transparent conductive polymer known as PEDOT:PSS rather than expensive silver nanowires or printed metal mesh.

Electronic nose. As interesting as it sounds, an electronic nose is basically an array of gas sensors with each having a slightly different function than the other. When exposed to a mixture of gases, there is a distinct relative response from each individual pixel. With proper software, a composite response can be worked out to disentangle the contributions from each of the pixels on this array.

Fig. 8 shows an electronic nose made from carbon nanotubes and

injected on a flexible polymer substrate. The inks on the bottom-right display real-time responses to methane, carbon dioxide, and ammonia concentrations, showing that the gas sensor is capable of effectively analysing a wide range of gases.

However, it cannot detect tiny traces of hazardous gases, mainly because it's not designed to identify specific gases at a molecular level. For common gases, it works quite well and can disentangle a mixture of gas concentrations.

As seen in Fig. 9, PragmatIC and Arm along with Manchester University have collaborated to produce an electronic nose that includes eight gas sensors based on organic transistors. The ICs used are much more cost-effective than silicon chips placed on a flexible plastic.

Progress made in 2021

Since printed flexible electronic circuits offer a wide range of benefits

such as flexibility, rapid prototyping, lower costs as compared to conventional electronic circuits, it is worth noting the progress made so far. Here's a brief rundown of their growth.

Healthcare and wellness. Advancing from electronic skin patches that have a rigid box to house electronic circuitry, Accensors (a German firm) has created flexible electronic skin patches that monitor temperature and cardiovascular parameters such as heart rate, ECG, and pH. Some microfluidics are added to monitor iron concentrations in sweat.

Quad Industries, in association with Henkel and Byteflies, has deployed skin patches in Belgian hospitals for epilepsy monitoring. Mekoprint has commercialised smart diapers that consist of electrodes made with carbon inks. On detecting moisture, the resistance changes, which is then processed